

Assignment 5: Product Development Process

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Part 1 — Project Topic

For my semester project I have selected the **yard tool interface** topic and refined its scope, since as listed it names an application domain rather than a problem. I narrowed it in three ways. The user population is manual wheelchair users with full upper-limb function: a powered chair already supplies much of the reaction mass and traction that make this problem difficult, and users with limited upper-limb function face a different governing problem, which is actuating the tool rather than reacting its forces. The task class is handheld powered tools — string trimmers, edgers, hedge trimmers and debris blowers — because ride-on equipment already has commercial adaptations and unpowered hand tools generate no meaningful reaction forces. The operating environment is residential turf, which is soft, uneven and frequently sloped. The resulting application space is **independent residential yard maintenance by manual wheelchair users with full upper-limb function, using handheld powered tools on unimproved ground.**

My initial impression is that handheld powered yard tools are designed around an assumption that stays invisible until it fails: a standing operator. A standing user reacts tool torque through a two-handed stance and body weight, repositions by stepping, and changes working height by bending or kneeling. A seated manual wheelchair user loses all three at once. The chair becomes the only available reaction mass, and it is a mass with a defined and fairly narrow stability envelope — the manual wheelchair lecture made the point that pitch stability is governed by axle position relative to the centre of mass, and that moving the axle forward to reduce rolling resistance directly reduces how hard it is to tip. The sharper constraint is that manual propulsion is itself a two-handed, cyclic task, so holding a tool occupies exactly the limbs that mobility requires and the user cannot propel and work at the same time. Terrain compounds both problems, since manual chairs propel poorly on grass and small casters catch on soft ground. The problem is therefore the simultaneous management of reaction forces, working envelope and mobility, and I suspect it is the coupling among the three, rather than any one of them, that has kept it unsolved.

I am deliberately not proposing a solution at this stage, since concept generation belongs in Phase 1 after needs are gathered. What I can identify are the areas where innovation is most likely to be required and where a prototype could be meaningfully tested. The first is the reaction-force path and the stability margin it leaves: how tool loads are routed into the chair without pushing the occupied system past its ISO 7176-1 static and ISO 7176-2 dynamic limits on a grade. The second is the seated reach and working envelope, which has to deliver a tool both to ground level and to a hedge at height from a fixed seated anthropometry. The third is mobility while equipped — how the tool is stowed or decoupled to return the hands to the handrims, and how it releases under fault. Each of these is mechanical, each is measurable, and each can be prototyped and tested in the spring using ramp tip-angle testing and instrumented load measurement. The scope is also bounded in the direction the Design Project Description asks for: this is an interface and mounting problem, not a new tool and not a new wheelchair.

Part 2 — Development Plan Based on the Product Development Process

What I would bring to my boss is a request to begin Phase 0, not a request to approve a design. Before releasing Phase 0 resources I would expect to provide six things: a defensible estimate of the served population, which is manual wheelchair users who own or maintain a yard and is a much smaller number than the wheelchair population generally; evidence that the need is real and unmet, gathered from interviews with at least eight manual wheelchair users who currently do yard work or have given it up, together with two or three occupational therapists or assistive technology professionals; a competitive and patent landscape covering adaptive gardening tools, tool-holding arms and wheelchair-mounted accessories, with a preliminary freedom-to-operate view; a regulatory determination; a funding model; and a resourced schedule with stated kill criteria. The regulatory and funding pair matters most, because it sets the economics before any engineering begins. The product makes no medical claim, so it is not an FDA-regulated device and is governed instead by power-tool safety standards and by ISO 7176 where it meets the chair. By the same token it will not satisfy a medical-necessity test, so there is no reimbursement path and the product must be cash-pay or nonprofit-subsidised. That fixes a price ceiling before a concept exists, and the price ceiling constrains the bill of materials.

Phase 0, Planning, is investigation, scoping and engineering assessment. Its deliverables are the validated needs list, the served-population estimate, the landscape report, the regulatory memo, a preliminary hazard list covering a powered cutting tool operated beside an occupied chair, and a resourced plan for Phase 1. The exit gate is an evidence gate, not a design gate: is the need demonstrated, is the population large enough to justify tooling, does a feasible stability envelope appear to exist, is there freedom to operate, and is the budget within tolerance. **Phase 1, Concept Development**, converts customer needs into a Product Design Specification and generates and screens concepts against it. The assistive-technology considerations from the lecture bear directly here — user-centred design, the accessibility ecosystem, and stigma and aesthetics — and the last of those deserves more weight than it would for a generic tool accessory, because a device that marks its user as needing help is a device that ends up left in the shed. Phase 1 exits with an approved PDS and a selected concept.

Phase 2, System-level Design, fixes the architecture: how tool loads are routed into the chair frame, where the interface attaches, and how the assembly mounts and demounts. Ergonomics and anthropometrics, adaptability across chair geometries, and safety and reliability are the governing concerns, and prototyping begins here. For this project the critical early experiment is a ramp tip-angle test under a representative tool load, because if the stability envelope cannot be met then the architecture is wrong and it is far cheaper to discover that in Phase 2 than later. **Phase 3, Detail Design**, resolves components, materials and tolerances, and is where compliance and standards, materials and hygiene, and the tension between customisation and scalability get settled. That tension is real for this product: wheelchairs vary enough in frame geometry that a single universal mount may not exist, and choosing between a family of fittings and one adjustable fitting is a Phase 3 decision with large manufacturing consequences.

Phase 4, Testing and Refinement, verifies the design against the PDS and validates it with users. Trials with the target population and long-term reliability testing are both necessary, but I would argue for extending the phase past conventional verification, because the endpoint that actually matters for assistive technology is continued voluntary use: a device can meet every specification and still be abandoned. **Phase 5, Production Ramp-up**, is small-batch

manufacture rather than mass production at this volume, with user training and a support and spares channel treated as deliverables rather than afterthoughts, since a user whose device fails has no substitute. Overall I would propose Phase 0 at roughly eight weeks of part-time effort with a formal gate before any Phase 1 money is committed. The argument for that discipline comes from the design reading: by the close of conceptual design the great majority of total product cost is already committed even though almost none of it has been spent, so a well-evidenced decision to stop is inexpensive now and very expensive later.

References

Dieter, G. E. and Schmid, L. C. *Engineering Design*, 5th ed. New York: McGraw-Hill, 2013. Chapter 2, Sections 2.1–2.4 and 2.6.

585.617 Rehabilitation Engineering, Module 5 lecture materials: "Product Development Process Overview" and "Considerations Specific to Assistive Technologies." Johns Hopkins University.

International Organization for Standardization. ISO 7176 series, *Wheelchairs* — Part 1 (determination of static stability) and Part 2 (determination of dynamic stability of electric wheelchairs). Geneva: ISO.